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Advanced Sequence Modeling Framework for Analyzing Learner Feedback in Educational Systems

K. Nirmala Devi[1,*], Radhika K[2], and Chandrasekaran Neelakandan[3]

[1]Associate Professor, Master of Computer Application, MITS.
[2]Assistant Professor, Master of Computer Applications, Madanapalle Institute of Technology & Science (Deemed to be University), India.
[3]Foreign Expat, School of Information Engineering, Shaoguan University, Guangdong, 521000, China.
[*]Corresponding Author: K. Nirmala Devi. Email: drnirmaladevik61@gmail.com

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ABSTRACT: Finding the opinion of the students improves the teaching learning process. It acts as information to change the teaching style of a lecturer. Sentiment analysis is used to analyze feedback of the students, which helps us to find their polarities, that is, positive, negative, and neutral. Sentiment analysis of the whole sentence focused on the sentiment of the highest confident word. Previous research shows that the sentiments alone are not enough to observe the feelings of the students. Different words express different sentiments in a sentence. There is a need to extract the targets in a given sentence, which helps to find the sentiment towards those targets. In this study, we proposed a novel model to find the targets of the given sentence using hybrid BILSTM BIGRU techniques. We evaluate the model using the Facebook dataset.

Keywords: Learner Feedback, Educational System, Sentiment Analysis, Deep Learning, Sequence Modelling.

1 Introduction

Students' feedback can highlight the different issues of the students about the lectures. They express their feelings or emotions about the lectures in the feedback. Students' feedback is the teaching-learning process, equipping both the teachers as well as the students. Using students' feedback lecturer changes the way of teaching [1]. Students' feedback mostly contains the emotions of the students about their course and the instructor. One of the best methods used to classify the emotions of the text is sentiment analysis. Sentiment analysis is used in various applications to review the feedback. It is also powerful when it is applied to a specific domain [2]. Sentiment analysis using deep learning gives better results compared to previous research. Deep learning automatically captures the features with less training and also consumes less time. In the proposed work, we use deep learning techniques to improve the result [3].

In our previous work, we found the polarity of the students, but that information alone is not enough to observe the feelings of the students. Our proposed model is used to observe the overall perception of the students in a clear manner. Our model contains three parts: i) Polarity detection: Find the polarity of the students from students' feedback using a multi-embedding and multi-head attention fusion model. ii)Target extraction: Find the targets from the given sentence using the Bi-Integrated CRF model. iii) Target-based sentiment analysis: Merging the sentiment model + target model to find the target and sentiments towards those targets. In this study, we focus only on the target extraction using the Bi-Integrated CRF model. Given a sentence as input, it identifies the target and returns a list containing all target terms. For example: I liked the staff, Mr. Joe, but not the subject Embedded system, the targets for this sentence are {staff, MrJoe, subject, Embedded system}. We evaluate the effectiveness of the approach using the dataset of the Computer Science department at the University of Michigan, which contains a Facebook student group and also from a survey with the students where students define their experience with classes. The final data set consists of 1,042 sentences written by both graduates and undergraduates, describing both classes and instructors [4]. We compare the proposed model with the results of the BILSTM CRF model. The proposed model gives better performance compared to the baseline model.

2 Related Works

Thinking of others has always been an important information. Student feedback is the information used to equip the teacher as well as the students. Previous research on students' feedback based on a specific question called a reflective prompt [5]. The opinion of the students was collected based on this prompt. Different aspects of the students cannot be observed through this method. Another method to obtain the feedback is to ask questions directly to the students. In this method, not all students will answer the questions [6]. Sentiment analysis is used to identify the emotions described in a text. Previous researches concentrate on e-learning [7]. [8] argues that students do not have the proper equipment to express their emotions. They also argue that emotion of the students plays an important role in learning because positive emotions influence the students to perform better, and negative emotions distract the students from learning. Previous methods have some challenges in sentiment analysis, including i) similar lexical features show different sentiments, ii) Different forms of writing but same sentiment, iii) subject name and professor can appear in different forms, iv) sentiment lexicons are not sufficient for sentiment analysis v) feedback may not be genuine [9][10]. In this study, we proposed a model which identifies the students' sentiment towards the courses and instructors as stated in their remarks. [28][29] Targets are words through which opinions are expressed. It is very important because without knowing the target, the opinion of the sentence has limited use.

Previous research based on target extraction has of limited results, and they are in need of additional resources. The authors extracted the opinion words and the targets using double propagation. Because they strongly believed that opinion words related to targets [11]. The work in [12] proved that rich pooling functions extract the feature vector for the targets [13]. Combined word embeddings with target embedding to learn the target in the sentence, and also used two networks for the right and left context. This approach gave me the idea of an integrated approach, through which we achieved better results [3]. There are several deep learning techniques that show promising results on sentence-level or document-level sentiment analysis. The advantages of the deep learning method are continuous learning of text. There are two steps in sentiment classification: i) learn the word vectors continuously from using an embedding layer, ii) learn these resultant vectors using deep learning techniques [14]. Sentiment analysis can be performed efficiently by implementing deep learning models, including LSTM [15], BILSTM-CRF [3][16], Dropout layers [17], Dense [18], BIGRU [19][20]. [3] BILSTM -CRF is used to extract the target from the given sentence and classify the sentence according to the targets, and find the sentiments of classified sentences. It gives better results compared to the previous methods [21]. Lample et al BILSTM CRF is the most effective method used to classify sentences from social media and removes noisy and short messages from social media messages [3]. Classifies the sentence in a better way, and it acts as a base for all sentiments [22]. LSTM classifies long sentences and gives better results compared with the traditional methods. [23] proved that the fusion model of BIGRU gives better results. [24] GRU learn the dependencies in lengths of considerably large sequences in a better way. [25] GRU is a more efficient method compared to LSTM and reduces the complexity of LSTM [26]. The dropout technique is used to reduce overfitting on the training data [17]. They showed that dropout applied in hidden layers gives better results. They also show that adding noise is not only suitable for unsupervised feature training but also suited for supervised training features.

3 Experiments

3.1 Dataset

Sentences are extracted from a Facebook student group and also from a survey with the students, where students define their experience with classes in the Computer Science department at the University of Michigan. The data set consists of 1,042 sentences written by both graduates and undergraduates, describing both classes and instructors. Sentences are annotated to identify the course and instructors. Sentiments are labelled as positive, negative and neutral. If there is no explicit sentiment and no evidence in the given sentence, then it is considered neutral. Table 1 shows the students' feedback with the annotation that contains the class ID, instructors and sentiment towards that statement are expressed. In total, the 1,042 sentences include 976 class mentions and 256 instructor mentions, for a total of 1,232 entities [27].

Table 1: Students' feedback on the annotation

S. No	Students feedback	Annotation
1	The labs in EECS 588 are very engaging	{class id=588,department=EECS, Sentiment=positive}
2	When I took 561, Rosita Williford was the professor	{class id=561,instructor=Rosita Williford, Sentiment=neutral}
3	I didn't like EECS 417	{class id=417,department=EECS, Sentiment=negative}

3.2 Target Extraction

Sentences are preprocessed using the padding technique so that the maximum length of sentences is fixed for the network. In which short sentences are padded with zero. In the proposed work, the maximum length of each word of each sentence is set as seventy (70). The architecture of the proposed work is given in Figure 2.

During training, sentence xt is given as an input to the Bi-Integrated model, which processes the input in two directions. In both, the input is first passed through the embedding layer, which converts the word vectors into to global word co-occurrence matrix using the Glove pretrained model. The size of these embeddings is 300. It finds the words that are closely related to the target using the probability ratio. The probability of the word vector depends on the probabilities of co-occurrence rather than itself. A co-occurrence matrix can be found using the formula below,

$$M = \sum_{i,j=1}^{v} f\left(Z_{ij}\right)\left(w_i \widetilde{w}_j + b_i + \tilde{b}_j - log(Z_{ij})\right)^2 \quad (1)$$

M is a co-occurrence matrix, $f(X_{ij})$ is a weighting function calculated using $f(x) = \begin{cases}(x/x_{max})^{\propto} \\ 1\end{cases}$ $w_i\widetilde{w_J}$ words i and j, $b_i\widetilde{b_J}$ bias is used to restore the symmetry $log(X_{ij})$ used to maintain the sparsity of X.

The output of the embeddings is fed as input to the BILSTM layer. [3][16] Bi bidirectional LSTM is used to find past and future context information. The input of the BILSTM is the sequence of word vectors from the embedding layer, and it is processed along with the hidden states. BILSTM works in both forward $\overrightarrow{h_t}$ and backward direction $\overleftarrow{h_t}$. The output of the BILSTM is the concatenation of both the LSTM, represented as y_t.

$$\overrightarrow{h_t} = \text{LSTM}\ (w_t, \overrightarrow{h_{t-1}}) \tag{2}$$

$$\overleftarrow{h_t} = \text{LSTM}\ (w_t, \overleftarrow{h_{t+1}}) \tag{3}$$

$$y_t = (\overrightarrow{h_t}\ ,\ \overleftarrow{h_t}) \tag{4}$$

The outputs of the BILSTM layers are scores for each target. The scores of the targets are given to the BiGRU layer. [19][20] BiGRU is a combination of two GRUs (forward and backwards), which is used to find the most relevant words using two gates: i) reset ignores the irrelevant words, and ii) update gate updates the current value with the previous value if the score is larger. BiGRU is the concatenation of forward $\overrightarrow{h_{t-1}}$ and backwards $\overleftarrow{h_{t+1}}$ represented using the following equation,

$$\overrightarrow{h_t} = \text{GRU}\ (w_t, \overrightarrow{h_{t-1}}) \tag{5}$$

$$\overrightarrow{h_t} = \text{GRU}\ (w_t, \overleftarrow{h_{t+1}}) \tag{6}$$

$$y_t = (\overrightarrow{h_t}, \overleftarrow{h_t}) \tag{7}$$

Dropout layers are added in both directions to remove the overfitting value after the BIGRU layer [17]. The dropout range for this model is given as 0.01. The output from both directions will be concatenated using the concatenation layer. The output of concatenation will be given to the dense layer [18]. A dense layer is a collection of neurons in a neural network. Each neuron in the dense layer receives input from all the neurons in the previous layer, so it is called a densely connected layer. The output of the dense layer is represented using the following equation,

$$\mathrm{O_t} = \mathrm{ReLU}\big((\mathrm{y_t} \times \mathrm{w_t}) + \mathrm{bias}\big) \tag{8}$$

Where O_t is the output of the dense layer, the weight matrix w_t created by the layer, bias vector created by the layer, and activation function ReLU passed as the activation argument. [3][16] Conditional random fields find the probability of the input sequence with the tags. Let x be the input and y be the tag, probability is computed using,

$$p(y/x) = \frac{\exp\ (score(x,y))}{\sum_{y'} \exp\ (score(x,y'))} \tag{9}$$

$$score(x,y) = \sum_{i=0}^{n} W_{y_{i-1},y_i} . LSTM(x_i) + b_{y_{i-1}y_i} \tag{10}$$

Where y' ranges over all possible output sequences, W is the weight vector, and b is the bias corresponding to the transition function y_{i-1} to y_i.

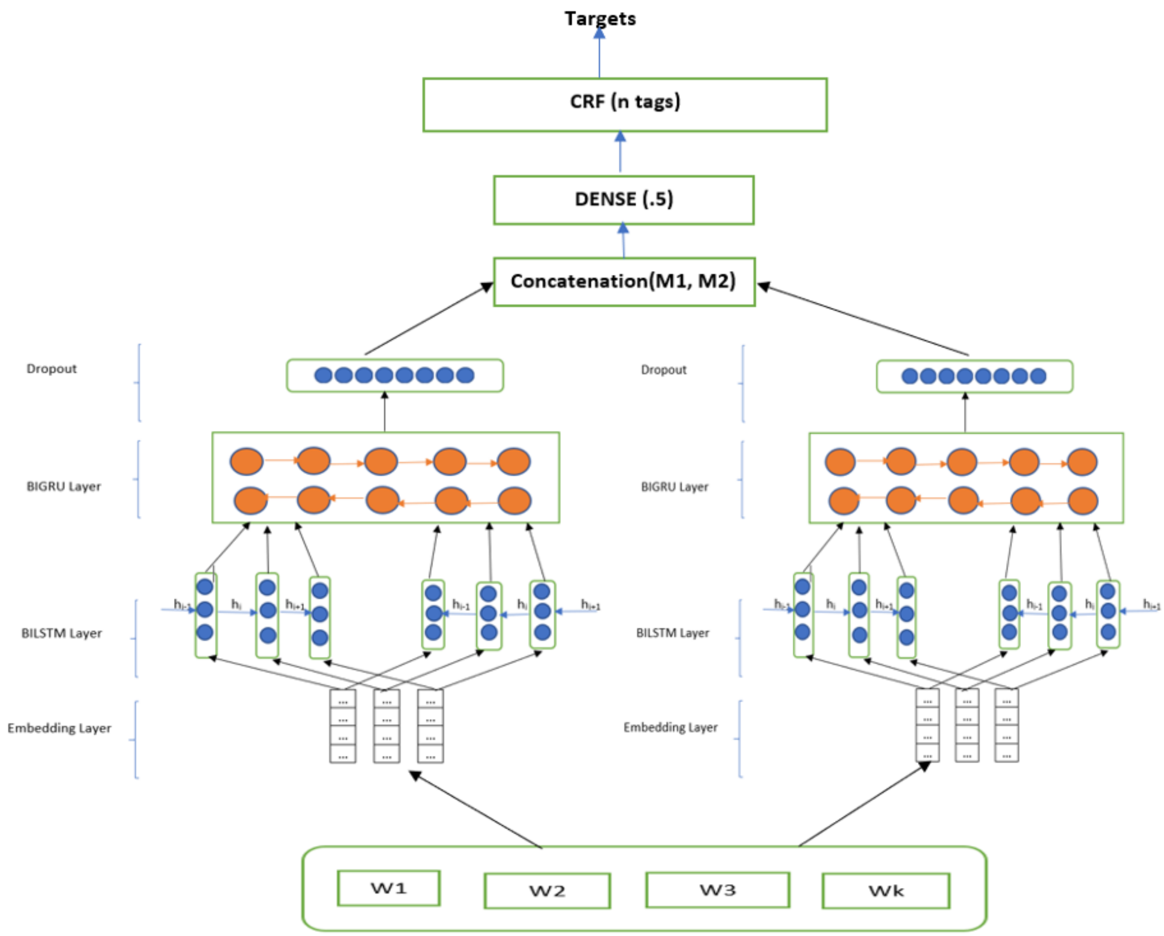

Figure 1: Bi-Integrated CRF Model

4 Results

4.1 Evaluation metrics

To measure the performance of the model, we used classification metrics that include Accuracy (AC), recall (RL), precision (P), F1 score (F1) and misclassification rate. [confusion matrix]. Performance of the model is evaluated using the count of test data correctly and incorrectly, using four evaluation metrics, including N_{TP}, N_{FP}, N_{TN}, N_{FN}. True Positive Rate (N_{TP}): Actual value is positive and predicted value is positive, False Positive Rate (N_{FP}): Actual value is positive and predicted value is negative, True negative Rate (N_{TN}): Actual value is negative and predicted value is negative, False negative Rate (N_{FN}): Actual value is negative and predicted value is positive. The standard evaluation metrics can be defined as in equations (11) to (14),

$$AC = \frac{N_{TP} + N_{TN}}{N_{TP} + N_{TN} + N_{FP} + N_{FN}} \quad (11)$$

$$P = \frac{N_{FP}}{N_{TN} + N_{FP}} \quad (12)$$

$$RL = \frac{N_{TP}}{N_{TP} + N_{FN}} \tag{13}$$

$$F_1 = 2 \times \frac{\mathrm{P} \times \mathrm{RL}}{\mathrm{P} + \mathrm{RL}} \tag{14}$$

4.2 Comparison with baseline model

For training and testing, we use the Facebook and students survey data set provided by Charles Welch, Rada Mihalcea 2016 [27]. It contains 1,042 sentences, including 976 class mentions and 256 instructor mentions, for a total of 1,232 entities. The data set has 2571 unique words. 70% of the data is used for training, and 30% is used for testing data. The input sentence is given to the embedding layer glove (pretrained model). The size of the embedding is 300. The 'rmsprop' optimization is used to train the network with a learning rate of 0.5. And after each training epoch, the network is tested on the validation data. For training the Bi-Integrated CRF model, ReLU is used as an activation function, a dropout rate of 0.4, and the size of the word vectors is 300. We compared the proposed model called the Bi-Integrated CRF model with the Bidirectional CRF evaluation criteria, including Accuracy (AC), Precision (P), Recall (RL), and F1Score (F1) given below:

Table 2: Performance comparison of Bi-Integrated CRF with Bi-Directional CRF

Model	Accuracy	Precision	Recall	F1Score
Bi-Integrated CRF	98.40%	77.57%	75.55%	75.98%
Bi-Directional CRF [47]	98.76%	82.73%	81.44%	81.57%

It is observed that, compared with the baseline model, Bi-integrated CRF gives better results. This indicates our approach outperforms the baseline model.

4.3 Comparison charts for evaluation metrics

The figure below compares the performance of the proposed approach, Bi-integrated CRF, with the existing method, Bidirectional CRF. Figure 2 shows the accuracy, Figure 3 shows the precision, Figure 4 shows the Recall, and Figure 5 shows the F1 score.

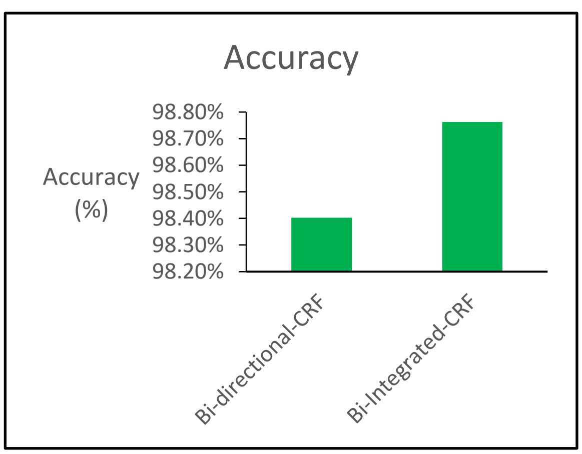

Figure 2: Comparison chart using Accuracy

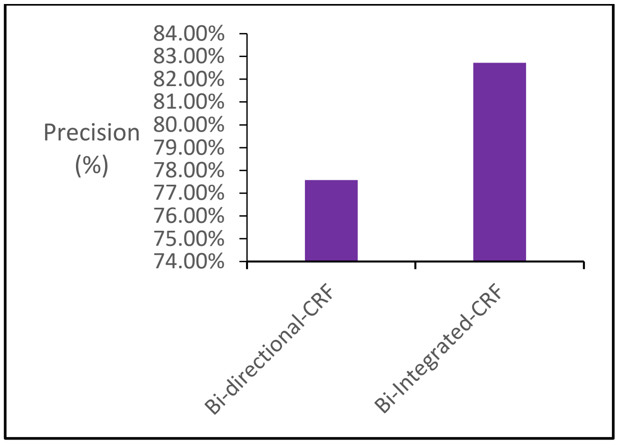

Figure 3: Comparison chart using Precision

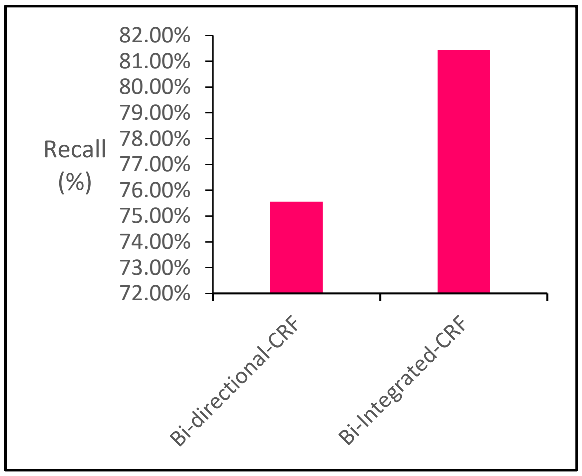

Figure 4: Comparison chart using Recall

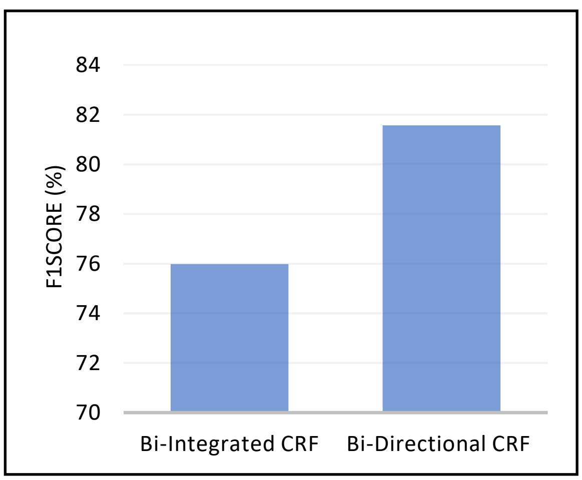

Figure 5: Comparison chart using F1 score - MISSING

5 Conclusion

This paper has presented a novel approach to improve the performance of the target extraction model. The approach employs a Bi-Integrated CRF to extract the targets in the students' feedback.

We have conducted experiments to evaluate the proposed model using the Facebook datasets [27] and compared it with the baseline model that employs Bidirectional-CRF [3]. Empirical results show that our approach achieves state-of-the-art performance on the baseline model. In future studies, we plan to explore the target for the given sentence and also find the sentiments for the same sentence. Using statistical analysis, the overall observation of students will be found.

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Conflicts of Interest: The authors declare no conflicts of interest to report regarding the present study.

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